Assignment #7 for Numerical Analysis

Deadline: Apr. 25

1. (40 points) Given data points $(0,0),(1,3),(2,3),(5,6)$.
   (a) Find the best line through the data points.
   (b) Find the best parabola through the data points.
   (c) Find the best degree 3 polynomial through the data points. Also, find the degree 3 interpolating polynomial, and compare.
2. (40 points)
   (a) Use Newton's divided differences to find the degree 2 interpolating polynomial $P_2(x)$ through the points $(0,0),(\pi/2,1)$, and $(\pi,0)$.
   (b) Calculate $P_2(\pi/4)$, an approximation for $\sin(\pi/4)$.
   (c) Use Theorem 3.4 to give an error bound for the approximation in part (b).
   (d) Using a calculator, compare the actual error to your error bound.
3. (20 points) Let $x_j (j = 0, 1, \cdots, n)$ be distinct nodes.
   (a) Show that $\sum_{j=0}^{n} x_j^k L_j(x) \equiv x^k \quad (k = 0, 1, \cdots, n)$.
   (b) Show that $\sum_{j=0}^{n} (x_j - x)^k L_j(x) \equiv 0 \quad (k = 1, 2, \cdots, n)$.